ANGEL DAVID NÚÑEZ PAREDES

SOFTWARE ENGINEER | PYTHON BACKEND DEVELOPER

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PROFESSIONAL SUMMARY

Software Engineer and Biomedical Engineer with a Master's degree in Electrical Engineering, specializing in Instrumentation. I specialize in developing backend applications and engineering software that integrate APIs, databases, automation, data processing, and modular architecture. I apply software development practices such as asynchronous processing, automated testing, containerization, and CI/CD to build maintainable, reliable, and reproducible solutions.

My engineering background allows me to approach software development from a multidisciplinary perspective, combining programming, electronics, and instrumentation. I am particularly interested in systems for acquiring, storing, processing, and analyzing data from sensors and instruments. My experience includes software development and integration, microcontroller programming, electronic design, experimental validation, equipment calibration, and technical documentation for advanced medical and instrumentation systems.

I enjoy solving complex technical problems, learning new technologies, and building clear, well-structured, testable software that addresses real-world needs.

TECHNICAL SKILLS

Programming Languages: Python, C, C#, SQL.
Backend Development: FastAPI, REST APIs, Pydantic, SQLAlchemy, Alembic, JWT/OAuth2 authentication, Role-Based Access Control (RBAC).
Databases and Data Processing: PostgreSQL, TimescaleDB, MySQL, SQLite, Redis, Pandas, NumPy, SciPy
Asynchronous Processing and Real-Time Communication: asyncio, Celery, Celery Beat, WebSockets.
Testing and Code Quality: Pytest, Ruff, MyPy, debugging, automated testing.
DevOps and Tools: Docker, Docker Compose, Git, GitHub Actions, CI/CD, Linux.
Frontend Development: TypeScript, React, Vite, TanStack Query, Tailwind CSS, Plotly.js.
Instrumentation and Embedded Systems: Data acquisition, signal processing, embedded systems, PIC and STM32 microcontrollers, biomedical instrumentation.

PROFESSIONAL EXPERIENCE

Institute of Applied Sciences and Technology (ICAT) – UNAM | Backend Developer – Automation and Data Applications

August 2025 – Present

- Develop software solutions in Python, C, and C# for engineering applications and technical environments.
- Implement data processing and analysis workflows, task automation, and information management solutions.
- Apply object-oriented programming, debugging, testing, validation, and technical documentation practices.
- Collaborate with multidisciplinary teams on research and instrumentation projects.

Immunoassay Center (CIE) | Biomedical and Technical Specialist – Medical Diagnostic Equipment

June 2023 – August 2024

- Program and test electronic systems for medical diagnostic equipment.
- Develop analysis and validation tools and partially automate measurement workflows and technical documentation.
- Contribute to research and development (R&D) projects by applying engineering and programming knowledge.

Center for Neuroscience of Cuba (CNEURO) | Biomedical Engineer – Scientific Reserve

February 2023 – May 2023

- Program PIC and STM32 microcontrollers in C to develop and validate biomedical solutions.
- Automate laboratory processes, analyze experimental data, and prepare technical documentation.
- Contribute to the development and validation of biomedical devices and technological solutions.
- Collaborate with multidisciplinary engineering and research teams.

SELECTED SOFTWARE PROJECTS

MedLab Platform — Laboratory Management Backend (LIMS)

GitHub: github.com/Davidn00/medlab-platform

Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Redis, Celery, Docker, Pytest, GitHub Actions

- Designed a layered, modular backend and versioned REST APIs featuring validation, pagination, filtering, and statistics.
- Implemented JWT/OAuth2 authentication and role-based access control, along with functionality for managing samples, test results, equipment, calibrations, maintenance, and audit trails.
- Implemented asynchronous processing, PDF report generation, and data export, alongside automated testing, logging, health checks, and CI/CD workflows.

InstrumentHub — Instrumentation Data Acquisition and Analysis Platform

GitHub: github.com/Davidn00/instrument_hub

Python, FastAPI, PostgreSQL/TimescaleDB, SQLAlchemy, Alembic, WebSockets, NumPy, SciPy, Pandas, TypeScript, React, Docker, Pytest

- Developed a modular architecture for acquiring, processing, storing, and analyzing data from instruments and sensors.
- Implemented APIs, signal processing components, and real-time communication, while developing the TypeScript and React frontend structure and configuring the local environment with Docker Compose.

EDUCATION

Master's Degree in Electrical Engineering – Instrumentation, National Autonomous University of Mexico (UNAM) | 2024–2026

Bachelor's Degree in Biomedical Engineering, José Antonio Echeverría Technological University of Havana (CUJAE) | 2018–2022

TECHNICAL PUBLICATION

"State-estimation as a virtual sensing scheme in a heat conduction experiment" — Latin American Congress on Automatic Control, October 2025.

DOI: 10.58571/CNCA.AMCA.2025.051

LANGUAGES

Spanish: Native

English: B2 (Upper-Intermediate)